$$= \lim_{\Delta x \to 0^+} \left(1 + \frac{1}{\Delta x}\right) \to \infty$$

Therefore $f'(0)$ does not exist.

Here we observe that the graph of f has a jump at $x = 0$. That is $x = 0$ is a jump discontinuity.

The above illustrations and examples can be summarised to have the following conclusions.

A function f is not differentiable at a point x_0 belonging to the domain of f if one of the following situations holds:

(i) f has a vertical tangent at x_0.

(ii) The graph of f comes to a point at x_0 (either a sharp edge $\vee$ or a sharp peak $\wedge$)

(iii) f is discontinuous at x_0.

A function fails to be differentiable under the following situations :

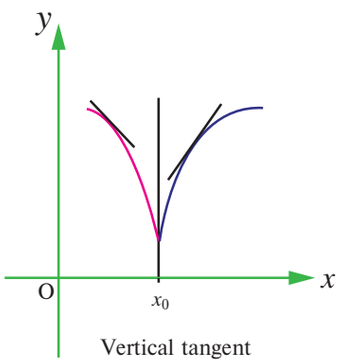


Fig. 10.21

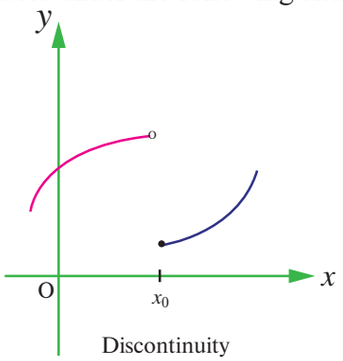


Fig. 10.22

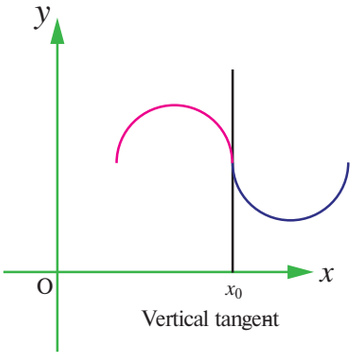


Fig. 10.23

We have seen in illustration 10.3 and 10.4, the function $f(x) = |x - 2|$ and $f(x) = x^{\frac{1}{3}}$ are respectively continuous at $x = 2$ and $x = 0$ but not differentiable there, whereas in Example 10.3 and Illustration 10.5, the functions $f(x) = \lfloor x \rfloor$ and $f(x) = \begin{cases} x & x \le 0 \\ 1 + x & x > 0 \end{cases}$ are respectively not continuous at any integer $x = n$ and $x = 0$ respectively and not differentiable too. The above argument can be condensed and encapsuled to state: **Discontinuity implies non-differentiability.**

Theorem 10.1 (Differentiability implies continuity)

If f is differentiable at a point $x = x_0$, then f is continuous at x_0.

Proof

Let $f(x)$ be a differentiable function on an interval (a, b) containing the point x_0. Then $f'(x_0) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$ exists, in the sense that $f'(x_0)$ is a unique real number.

Now $\lim_{\Delta x \to 0} [f(x_0 + \Delta x) - f(x_0)] = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \times \Delta x$

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